ABHINAY MUSKU

Hyderabad / Chennai, India | +91 9398707418 | abhinay.musku.reddy@gmail.com
linkedin.com/in/abhinay-musku | github.com/NOiD-777

PROFESSIONAL SUMMARY

Full-Stack & AI Engineer with hands-on production experience architecting scalable data pipelines and backend microservices using Python, PostgreSQL, SQL, Databricks, and PySpark across Microsoft Azure and AWS. Proven expertise building production Generative AI applications integrated with LLM APIs (Google Gemini API, OpenAI), LangChain, React, TypeScript, Node.js, and FastAPI. Strong track record in optimizing machine learning inference runtime and deploying durable agentic systems.

TECHNICAL SKILLS

- **Generative AI & LLMs:** LLM APIs (Google Gemini API, OpenAI), LangChain, Prompt Engineering, RAG Architectures, AI Agents, Hugging Face Transformers
- **Full-Stack & Web:** React.js, TypeScript, Node.js, Express.js, Angular, JavaScript, HTML5, CSS3, Tailwind CSS
- **Languages & Backend:** Python, SQL, PostgreSQL, FastAPI, Java, C#, C, RESTful APIs, Microservices
- **Cloud & Big Data:** Databricks, PySpark, Delta Lake, Microsoft Azure (Azure Services, Azure DevOps), AWS (EC2, ECS, Lambda), GCP
- **Machine Learning & Vision:** Scikit-learn, OpenCV, TensorFlow, Keras, NLTK, Pandas, NumPy, Matplotlib
- **Developer Tools & DBs:** PostgreSQL, MongoDB / MongoDB Atlas, Git, GitHub, Vite, Postman, Google Colab

PROFESSIONAL EXPERIENCE

JMAN Group | Chennai, India *(Feb 2026 – Present)*

Data Engineer — Full-Time (May 2026 – Present)

- **Enterprise Data Engineering:** Developing and maintaining scalable enterprise data pipelines using Databricks, PySpark, SQL, and PostgreSQL for financial services clients, optimizing high-volume query throughput and batch ETL reliability.
- **Cloud & API Ingestion:** Architected resilient API ingestion pipelines connecting Azure services and AWS (EC2, ECS, Lambda) to extract, transform, and store raw financial data streams in Delta Lake.

Software Engineer Intern (Feb 2026 – Apr 2026)

- **Cross-Functional Rotation:** Completed engineering rotation across Full-Stack Development, Data Science & AI, Power BI, Data Engineering, and Platform Engineering, building proof-of-concept components across core company services.

AI AXISS INC | Hyderabad, India *(Oct 2025 – Jan 2026)*

Angular Developer Intern

- **Frontend & API Integration:** Developed responsive, component-driven web applications using modern Angular, TypeScript, HTML5, and CSS3, integrating RESTful APIs for real-time data visualization and client-server synchronization.

Brihaspathi Technologies Limited | Hyderabad, India *(Aug 2025 – Sep 2025)*

Software Developer Intern — AI & ML

- **Computer Vision Edge Deployment:** Engineered and deployed a lightweight Face Recognition solution using FastAPI, Python, and OpenCV, achieving ~92% detection accuracy on low-resource edge hardware.

- **Performance Optimization:** Reduced CPU usage by 35% through custom skip-frame processing and image downscaling pipelines without compromising recognition reliability.

PROJECTS

ArtYatra — Full-Stack AI Cultural Platform *(Aug – Sep 2025)*

Stack: React, TypeScript, Node.js, Express.js, MongoDB Atlas, PostgreSQL, LLM APIs (Gemini API), LangChain

- **LLM APIs & Prompt Orchestration:** Engineered a scalable full-stack application leveraging LLM APIs (Google Gemini API) with structured prompt orchestration to analyze and classify 8 traditional Indian art styles with confidence scoring.
- **Contextual Retrieval & Backend:** Integrated LangChain prompt templates, TypeScript backend services, and database retrieval (MongoDB Atlas & PostgreSQL) to ground art history responses with sub-second response times.

Twitter Sentiment Analysis using ML *(Mar – Jun 2025)*

Stack: Python, Pandas, NLTK, Scikit-learn, Tweepy, Google Colab

- **Large-Scale NLP Pipeline:** Formulated and optimized a sentiment classification pipeline over the Sentiment140 dataset of 1.6M tweets, achieving a 25% accuracy gain through hyperparameter tuning and refined tokenization/stemming.
- **Model Calibration:** Benchmarked and calibrated multiple classifiers (Logistic Regression, Naive Bayes), reducing false positive predictions by 20%.

EDUCATION & LEADERSHIP

MLR Institute of Technology, Hyderabad, India — Bachelor of Engineering in Information Technology (CGPA: 8.18/10) | Nov 2022 – Apr 2026

- **Leadership & Community:** Technical Lead at Google Developers Group (GDG) on Campus MLRIT.
- **Hackathon Winner:** Winner – BuildAI4India Hackathon, Swecha Telangana (Aug 2025) for AI-powered cultural art preservation.
- **Certifications:** Smart Coder DSA Certification (Silver) - Smart Interviews (2025) | Google Cloud Computing Foundations - NPTEL (2024).